

Professional Experience

- Aug 2026–now **AI Research Engineer**, *Nu:legal*, Germany
- Built the agent harness for a legal AI platform, including tool integration and context management for reliable multi-step workflows.
- Feb–Jul 2026 **Data Scientist — Generative AI**, *OncoBrain*, USA
- Engineered a retrieval evaluation framework benchmarking full-text, vector, hybrid, and reranked-hybrid search strategies across quality and latency metrics, enabling data-driven selection of retrieval pipelines.
 - Built and maintained ETL pipelines ingesting multiple oncology guideline sources, resolving integration and runtime failures to improve data reliability and metadata quality.
- 2023–2025 **Teaching Assistant**, *Free University of Berlin*, Germany
- Taught “Programming Foundations for Psychological Science” to undergraduate psychology students.
 - Designed course materials covering object-oriented programming and core data-science libraries in Python.
- 2020–2025 **Research Scientist**, *Free University of Berlin*, Germany
- Conducted deep-learning research on human object vision using CNNs, time-series EEG, and ultra-high-resolution neuroimaging, resulting in two peer-reviewed publications.
 - Developed scalable, HPC-based processing pipelines for large MRI datasets using Python and Linux clusters.
 - Investigated tokenization approaches in LLMs to improve transfer learning for sentiment classification.
 - Led international collaborations across academic institutions, communicating complex modeling results to interdisciplinary teams.
 - Supervised one intern on modeling and data-analysis techniques.

Education

- 2020–2025 **PhD in Computational Neuroscience**, *Free University of Berlin*, Germany
- 2017–2019 **MSc in Neuroscience (Joint)**, *Universities of Göttingen and Bordeaux*, Germany / France
- 2010–2015 **BSc in Biology**, *University of Havana*, Cuba, *Summa cum laude*

Skills

- AI Engineering Python, PyTorch, Pandas, NumPy, Scikit-learn, LanceDB, LangChain, LangGraph, Git, FastAPI, SQL, Claude Code
- AI Modeling LLMs, RAG, Vision Transformers, Representational learning, Feature extraction, Classification, Regression, A/B testing, Fine-tuning, XGBoost, End-to-end training and validation, CI/CD pipelines
- Languages Spanish (Native), English (Fluent), German (Fluent)

Awards, Grants and Scholarships

- 2024 Einstein Center for Neurosciences Graduation Scholarship — 10,800 €
- 2019–2022 Einstein Center for Neurosciences PhD Fellowship — 63,000 €
- 2017–2019 Erasmus Mundus Joint Master Degree Scholarship — 49,000 €

Publications

- Peer-reviewed **Carricarte T** et al. (2026). Layer-specific spatiotemporal dynamics of feedforward and feedback in human visual object perception. *eLife*.
- Peer-reviewed **Carricarte T** et al. (2024). Laminar dissociation of feedforward and feedback in high-level ventral visual cortex during imagery and perception. *iScience*.